

R-11 Analysis: Constraint 1 Parity — Status and Fix Path

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R-11: Constraint 1 — Parity Analysis

1. The Problem

Paper #75 Section 4.3, Constraint 1 states:

“The short root multiplicity $m_s = 3$ is odd. The local Arthur parameter at the archimedean place determines the signature of the intertwining operator $M(\psi_{\text{inf}})$. For non-tempered parameters with even $SL(2)$ dimension d_i contributing to the short root, the sign $\epsilon(\psi_{\text{inf}}) = (-1)^{m_s - 1} = +1$ is inconsistent with the required $\epsilon = -1$ from the global root number. This eliminates 34 of 45 types.”

Problems: 1. The formula $\epsilon(\psi_{\text{inf}}) = (-1)^{m_s - 1}$ has no citation 2. It doesn't appear in Arthur [Art13] 3. The phrase “contributing to the short root” is vague 4. The “required $\epsilon = -1$ from the global root number” is not derived

2. What the Constraint Should Be

Constraint 1 should be one of the following established results:

Option A: The Adams-Johnson Inner Form Condition

For the inner form $SO(5,2)$ of the quasisplit $SO(7)$, the local archimedean A-packet $A_{\psi}(SO(5,2))$ may be empty for certain non-tempered parameters. The Adams-Johnson theorem [AJ87] classifies which Arthur parameters have nonempty local packets at each real form.

The condition involves the interaction between the $SL(2)$ factors $S_{\{d_i\}}$ and the signature $(5,2)$ of the real form. Specifically, the Hodge decomposition of the parameter must be compatible with the compact/noncompact structure of $SO(5,2)$.

Status: This IS a rigorous and established result. But the exact count of excluded types for $SO(5,2)$ with $\sum n_i d_i = 7$ requires explicit computation (45 cases). I

have NOT completed this computation.

Citation: Adams-Johnson, “Endoscopic groups and packets of nontempered representations,” *Compositio Math.* 64 (1987), 271-309.

Option B: Arthur’s Epsilon Sign Condition

Arthur [Art13, Theorem 1.5.1] defines the character ϵ_{ψ} on the centralizer S_{ψ} . A representation π in the A-packet contributes to the discrete spectrum only if $\epsilon_{\psi}(s_{\pi}) = 1$ for specific elements s in S_{ψ} .

For the inner form $SO(5,2)$, there is an additional Kottwitz sign:

$$e(SO(5,2)) = (-1)^{(q-1)/2} = (-1)^{2*1/2} = -1$$

The condition becomes: $\epsilon_{\psi} * e(G) = 1$, i.e., $\epsilon_{\psi} = -1$ for $SO(5,2)$.

For non-tempered parameters, the epsilon character depends on: - Local root numbers $\epsilon(1/2, \mu_i \times \mu_j)$ at each place - The $SL(2)$ dimensions d_i (which affect the sign through the L-function functional equation)

Status: This IS the correct framework, but computing ϵ_{ψ} for each of the 45 types requires evaluating root numbers at the archimedean place. This is doable but involves the Langlands-Shahidi method and is ~2 pages per type (or automatable via Atlas software).

Citation: Arthur [Art13], Theorem 1.5.1 and Chapter 6.

Option C: Computational Verification via Atlas

The Atlas of Lie Groups project (atlas.math.umd.edu, Adams-Vogan et al.) can compute Arthur packets for specific real forms. For $SO(5,2)$:

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atlas> set G = SO(5,2)
atlas> for each Arthur parameter psi with sum n_i d_i = 7:
atlas>   compute A_psi(G)
atlas>   check if A_psi contains discrete series
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This would give the EXACT count of excluded types.

Status: Requires running the Atlas software. Cannot be done from first principles in a CI session.

3. Discrepancy: 34/45 vs Table

The paper’s Appendix A table shows Constraint 1 in the “Killed by” column for ALL types except Type 3 (which suggests 44/45 killed, not 34/45). But the text says “34 of 45.”

Possible resolutions: 1. The table’s compressed entries (types 9-45) include individual types not killed by Constraint 1, but the group listing only shows the most

common constraints 2. The 34/45 count in the text is correct and the table is misleading 3. The 34/45 count is wrong

This discrepancy must be resolved. If 44/45 types are killed by Constraint 1 (as the table suggests), then combined with Constraint 3 killing Type 3 ($n_i = 6 > 4$), ALL 45 types are eliminated without Constraint 2. This would make R-9 completely moot.

4. Why R-11 is Essential

If R-9 (spectral gap) is dropped (because the 91.1 bound is wrong for $SO(5,2)$): - Without Constraint 1 AND without Constraint 2: types with ($n_i \leq 4, d_i = 2$) survive all remaining constraints {3, 4, 5, 6, 7} - These $d_i = 2$ types are NOT killed by Constraint 6 (Ramanujan at finite places — the depth $(d-1)/2 = 1/2$ matches the trivial bound) - They're not killed by Constraint 3 ($n_i \leq 4$) - Constraints 4, 5, 7 don't give hard eliminations

So WITHOUT Constraints {1, 2}: the proof fails for types with small (n_i, d_i). WITH Constraint 1 but without 2: the proof works (R-9 is moot).

Bottom line: R-11 is the ESSENTIAL fix for the elimination argument.

5. Recommended Path

1. Immediate: Replace the ad hoc formula with a citation to Adams-Johnson [AJ87] and Arthur [Art13, Theorem 1.5.1]. State: "By the Adams-Johnson inner form condition and Arthur's epsilon character, the non-tempered types compatible with the inner form $SO(5,2)$ are restricted to those whose $SL(2)$ parameters are (p,q) -compatible and whose epsilon characters match the Kottwitz sign $e(SO(5,2)) = -1$."
2. Verification: Run the Atlas of Lie Groups software to compute the exact count of excluded types. This is a finite computation (45 cases for a fixed real form).
3. Fallback: If the Atlas computation shows Constraint 1 kills FEWER than 34 types, check whether the remaining types are killed by Constraints {3, 5, 6}. If not, the paper needs Constraint 2 (R-9) after all.
4. Sarnak question: Ask Sarnak specifically: "For the inner form $SO_0(5,2)$ of the quasisplit $SO(7)$, what fraction of the 45 non-tempered Arthur types (sum $n_i d_i = 7$) contribute to the discrete spectrum? We claim 'at most 11 of 45' — is this consistent with the Adams-Johnson classification?"

6. Draft Replacement Text for Paper #75

Replace the current Constraint 1 paragraph with:

Constraint 1 (Inner form restriction). The real form $SO_0(5,2)$ is an inner form of the quasisplit $SO(7)$, with Witt index $q = 2$ and Kottwitz

$\text{sign } e(G) = (-1)^{\{q(q-1)/2\}} = -1$. By Adams-Johnson [AJ87] and Arthur [Art13, Theorem 1.5.1], a non-tempered Arthur parameter ψ contributes to the discrete spectrum of $\text{SO}(5,2)$ only if:

- (i) The local A-packet $A_{\{\psi_{\text{inf}}\}}(\text{SO}(5,2))$ at the archimedean place is nonempty (Adams-Johnson compatibility: the $\text{SL}(2)$ parameters $\{d_i\}$ must be compatible with the signature $(5,2)$), and
- (ii) The global epsilon character satisfies $\epsilon_{\psi} = e(G) = -1$ (Arthur's sign condition).

Condition (i) excludes parameters whose $\text{SL}(2)$ components are too large for the noncompact dimension $q = 2$ (e.g., S_7 , which requires $q \geq 4$). Condition (ii) imposes a parity constraint depending on the root numbers of the GL components $\{\mu_i\}$. Together, these eliminate [N] of 45 non-tempered types. [N to be verified by Atlas computation.]

Lyra, May 5, 2026. R-11 analysis. The fix is tractable (correct citations + finite computation) but requires either Atlas software or expert verification for the exact count.